

EXPERIMENT-2

AIM: To study about different types of network cables.

Different types of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial Cable
4. Fibre Optic Cable.

Cable Type: UTP

Category	Max. Data Transmission
Category 3	10 bps
Category 5	Up to 100 Mbps
Category 5e	1 Gbps



Advantages: Cheaper in cost

Easy to install as they have a small overall diameter.

Disadvantages: More prone to Electromagnetic Interference and noise.

Application / Use: 10-Base-T Ethernet
Fast Ethernet, Gigabit Ethernet.

Cable type: STP

Category	Max. Data Transmission
Category 6, 6a	10 Gbps
Category 7	10 Gbps.



Advantages: Shielded

Faster than UTP

less susceptible to noise and interference.

Disadvantages: Expensive

Greater Installation effort.

Application / Use:

Category 6/6a $\Rightarrow$ Gigabit Ethernet, 10G Ethernet (55 m) Widely used in data centres.

Category 7 $\Rightarrow$ Gigabit Ethernet, 10G Ethernet (100m)

Cable Type: Coaxial Cable

Category: RG-6, RG-59, RG-11

Max. data Transmission: 10-100 Mbps



Advantages:

- High bandwidth
- Immune to interference
- Low loss bandwidth
- Versatile

Disadvantages:

- Limited distance
- Cost

• Size is bulky

Cable Type: Fibre Optics Cable

Category:

- Single Mode
- Multimode

Max. Data Transmission: 100 Gbps

Advantages:

- High Speed & bandwidth
- High Security

Disadvantages:

- Expensive
- Requires skilled installers.

Application / use:

Maximum distance of fibre optics cable is around 100 metres.

RESULT:

Thus the ~~study~~ study about different types of network cables are executed successfully.